

Gut flora metabolism of phosphatidylcholine promotes cardiovascular disease

Metabolomics studies hold promise for discovery of pathways linked to disease processes. Cardiovascular disease (CVD) represents the leading cause of death and morbidity worldwide. A metabolomics approach was used to generate unbiased small molecule metabolic profiles in plasma that predict risk for CVD. Three metabolites of the dietary lipid phosphatidylcholine, namely choline, trimethylamine N-oxide (TMAO), and betaine, were identified and then shown to predict risk for CVD in an independent large clinical cohort. Dietary supplementation of mice with choline, TMAO or betaine promoted up-regulation of multiple macrophage scavenger receptors linked to atherosclerosis, and supplementation with choline or TMAO promoted atherosclerosis. Studies using germ-free mice confirmed a critical role for dietary choline and gut flora in TMAO production, augmented macrophage cholesterol accumulation and foam cell formation. Suppression of intestinal microflora in atherosclerosis-prone mice inhibited dietary choline-enhanced atherosclerosis. Genetic variations controlling expression of flavin monooxygenases (FMOs), an enzymatic source of TMAO, segregated with atherosclerosis in hyperlipidemic mice. Discovery of a relationship between gut flora-dependent metabolism of dietary phosphatidylcholine and CVD pathogenesis provides opportunities for development of both novel diagnostic tests and therapeutic approaches for atherosclerotic heart disease.